

Object Oriented Programming

Project Report:

ONLINE VOTING SYSTEM DATABASE

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Introduction:

This project implements a Online Voting Management System using Java Swing for the graphical user interface (GUI) and Oracle database for data storage. The system allows voters to sign up, log in, and access various functionalities related to the voting process, such as casting votes and viewing voter information.

Technologies Used:

- **Programming Language:** Java
- **GUI Library:** Swing
- **Database:** Oracle

Key Components and Methods:

Headers and Imports:

```
import javax.swing.*;  
  
import java.awt.*;  
  
import java.awt.event.ActionEvent;  
  
import java.awt.event.ActionListener;  
  
import java.sql.*;  
  
import java.text.ParseException;  
  
import java.text.SimpleDateFormat;
```

Attributes

```
private JFrame frame;
```

```
private int currentVoterId = -1;

private String district = "YourDistrict";

private int nextVoterId;

private Color lightBlueColor = new Color(173, 216, 230);
```

Methods

- **Constructor:** Initializes the main frame and sets up the initial UI.

```
public Main() {
    // Initialization logic...
}
```

- **loadImageIcon:** Loads and resizes the background image.

```
private ImageIcon loadImageIcon(String filename, int width, int height) {
    // Image loading logic...
}
```

- **initializeNextVoterId:** Fetches the next voter ID from the database.

```
private int initializeNextVoterId() {
    // Database interaction logic...
}
```

- **setCurrentVoterId:** Sets the current voter ID after successful login

```
public void setCurrentVoterId(int currentVoterId) {
    // Setter logic...
}
```

- **clearFrame:** Clears the frame content

```
private void clearFrame() {
    // Clear frame logic...
}
```

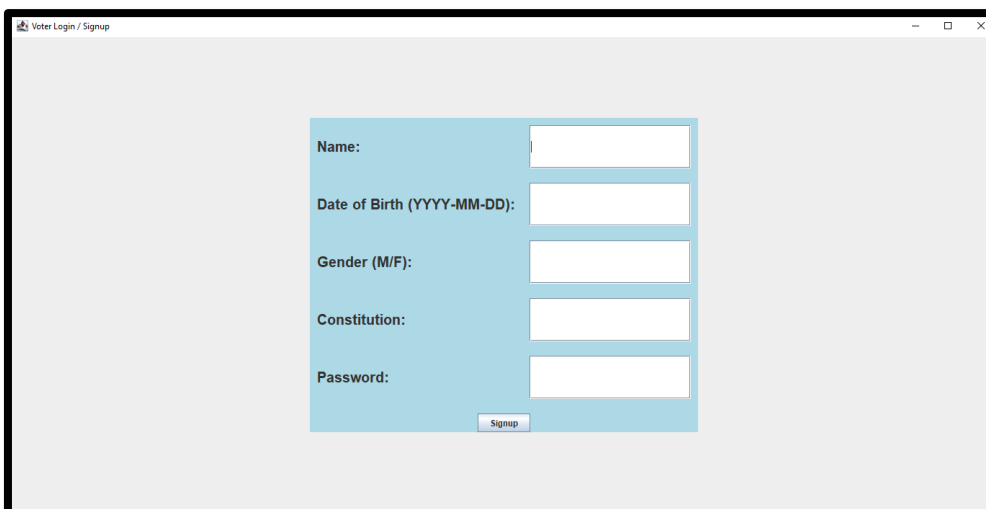
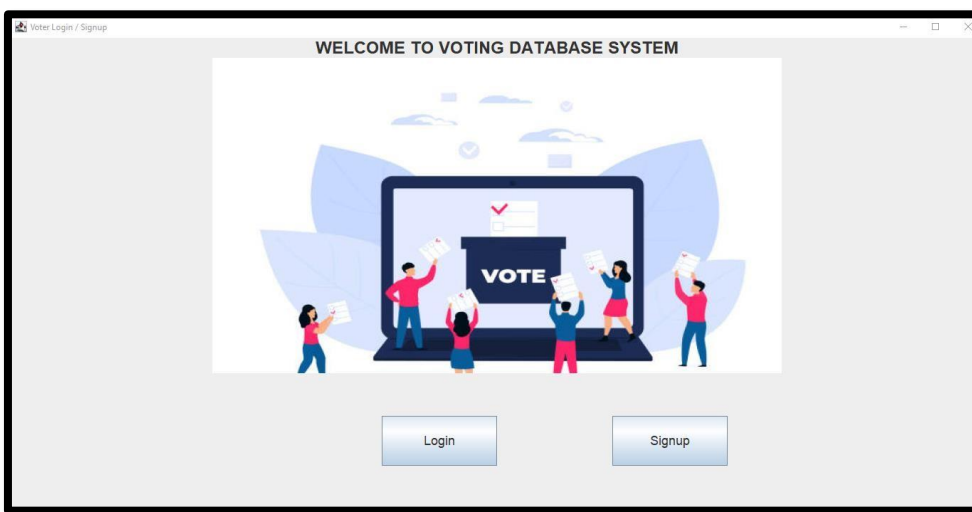
- **showVoteButtons:** Displays the main menu with options for casting votes and other actions.

```
public void showVoteButtons() {
    // Show vote buttons logic...
}
```

User Interaction Flow

1. **Initial Screen:** Users are greeted with a welcome message and options to login or signup.
2. **Login:** Users enter their Voter ID and password to login.
3. **Signup:** New users can create an account by entering their details.
4. **Main Menu:** Logged-in users can cast votes, view voter information, or access other functionalities.

- **OUTPUT SCREENS**

A screenshot of a web application window titled "Voter Login / Signup". The main content area displays a light blue rectangular form with the following fields and labels: "Name:", "Date of Birth (YYYY-MM-DD):", "Gender (M/F):", "Constitution:", and "Password:". Each field is represented by a white input box. A "Signup" button is located at the bottom right of the form.

Voter Login / Signup

Voter ID:

Password:

Login

Voter Login / Signup

Cast Vote

Admins

Candidates

Votes Information

Voter Info

Exit

Vote Form

Candidate: 12 - Pakistan Peoples Party

Vote ID: 9

Voter ID: 9

District: RAWALPINDI

Submit

Vote Form

Candidate: 12 - Pakistan Peoples Party

Vote ID:

Voter ID:

District: RAWALPINDI

Submit

Message

Vote inserted successfully!

OK

VOTES_ID	CANDIDATE_ID	VOTER_ID	DISTRICT	CAST_TIME
8	12	9	RAWALPINDI	2024-06-24 10:37:21.213
9	12	9	RAWALPINDI	2024-06-24 11:28:47.514
1	12	1	Lahore	2024-06-24 10:25:12.651
2	284	2	Islamabad	2024-06-24 10:25:12.656
3	232	3	Karachi	2024-06-24 10:25:12.662
4	136	4	Quetta	2024-06-24 10:25:12.666
5	174	5	Peshawar	2024-06-24 10:25:12.67
6	354	6	Faisalabad	2024-06-24 10:25:12.674
7	54	7	Multan	2024-06-24 10:25:12.679

Conclusion

This project demonstrates a simple yet functional online voting system with a user-friendly interface. The use of Java Swing for the GUI and Oracle for the database backend showcases effective integration of frontend and backend technologies in a desktop application.